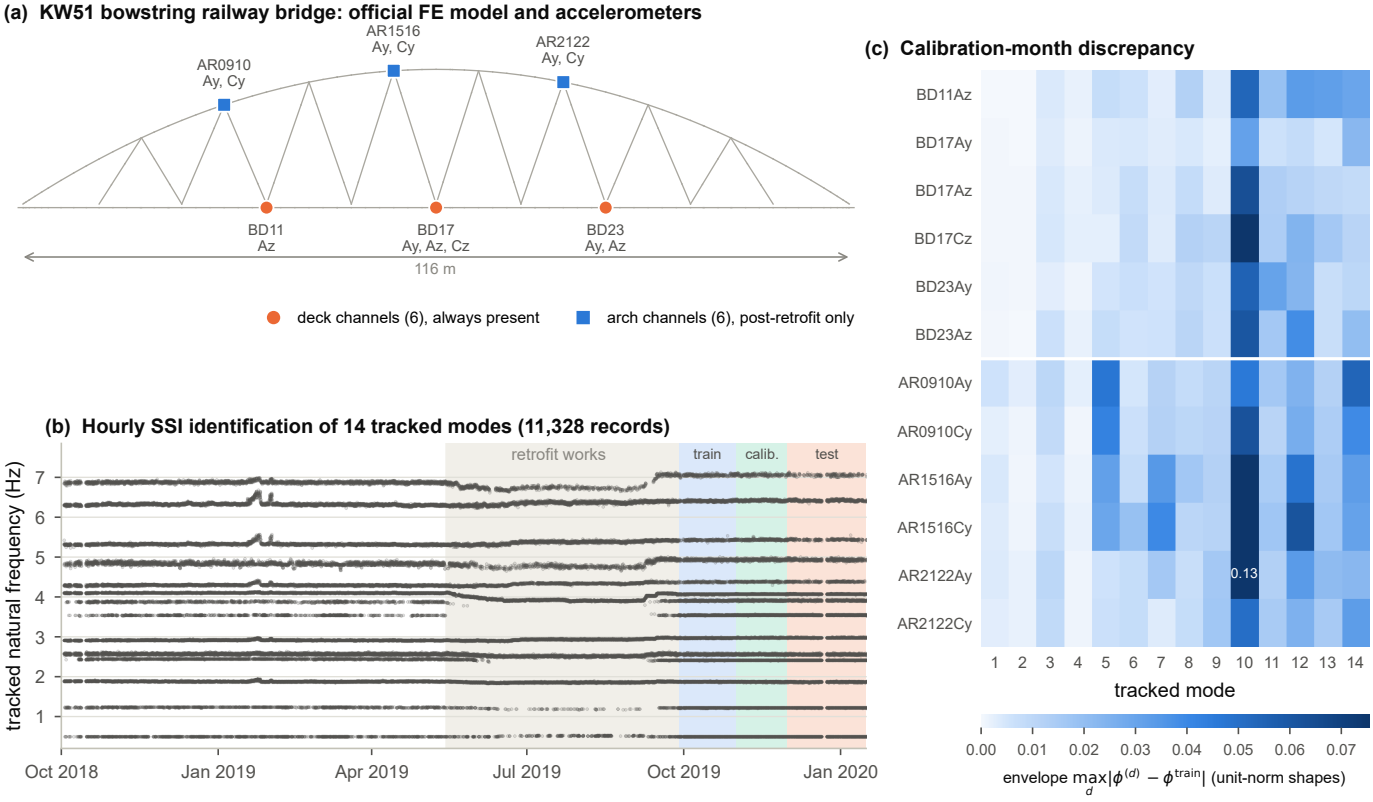

(a) KW51 bowstring railway bridge: official FE model and accelerometers
AR0910
Ay, Cy
AR1516
Ay, Cy
AR2122
Ay, Cy
BD11
Az
BD17
Ay, Az, Cz
BD23
Ay, Az
116 m
deck channels (6), always present
arch channels (6), post-retrofit only
(b) Hourly SSI identification of 14 tracked modes (11,328 records)
retrofit works
train
calib.
test
tracked natural frequency (Hz)
1
2
3
4
5
6
7
Oct 2018
Jan 2019
Apr 2019
Jul 2019
Oct 2019
Jan 2020
(c) Calibration-month discrepancy
BD11Az
BD17Ay
BD17Az
BD17Cz
BD23Ay
BD23Az
AR0910Ay
AR0910Cy
AR1516Ay
AR1516Cy
AR2122Ay
AR2122Cy
0.13
1
2
3
4
5
6
7
8
9
10
11
12
13
14
tracked mode
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
envelope $\max_d |\phi^{(d)} - \phi^{\text{train}}|$ (unit-norm shapes)